

**Integrative Multi-omic Approaches to Characterization of Crohn's Disease Immune
Profiles in Peripheral Blood Mononuclear Cells**

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ABSTRACT

Inflammatory Bowel Disease (IBD), particularly Crohn's Disease, presents unique challenges concerning immune dysregulation. This study aimed to explore whether peripheral blood mononuclear cells (PBMCs) exhibit distinct transcription factor binding and gene expression profiles in IBD. The dataset selected contains multi-omics data, including snATAC-seq and RNAseq, from two sample pools: Pool 3 (three healthy individuals) and Pool 4 (two Crohn's disease patients). snATAC-seq data quality was ensured through rigorous filtering and quality analysis, and subsequent snRNA-seq quality control and analyses were conducted using Seurat v5.2.1 to identify differential expression and motif enrichment. In tandem, these analyses aim to reveal where high genetic expression and immune activity correspond in the genome. PCA and UMAP were employed for dimensionality reduction and visualization, while gseGO facilitated pathway enrichment analysis. Our results highlighted differences in gene expression, particularly in immune and cell cycle pathways between the two cohorts. However, it is noted that at first glance there are no notable differences in gene expression of genes associated with Crohn's disease. In addition, the control group had enriched expression in pathways related to immune activation, whereas the case group had enrichment in cell cycle progression gene sets. This work provides the preliminary steps for further analyses, such as transcription factor footprinting or bind site region analysis, to elucidate genetic and epigenetic profiles of PBMCs in IBD. Ultimately, our study enhances the understanding of IBD pathogenesis by revealing potential genetic and epigenetic targets linked to immune dysregulation observed in PBMCs of Crohn's disease patients.

INTRODUCTION

Inflammatory bowel disease (IBD) encompasses a group of chronic inflammatory conditions of the gastrointestinal tract, affecting approximately 5 million people globally. Central to the pathogenesis of IBD is the dysregulation of immune responses, particularly within T cells. T cell migration and activity contribute significantly to gastrointestinal inflammation, making them a promising target for therapeutic intervention (Marsal et al., 2012). Despite their clinical significance, the precise epigenomic inflammatory signatures of T cells in IBD are not fully understood. Many genome-wide association studies (GWAS) have analyzed genetic associations of Crohn's disease specifically in intestinal tract cells. However, there is a component that many studies neglect: the peripheral blood that contributes immune cells and plays a role in inflammatory responses (Brown et al., 2024; de Lange et al., 2024; Yeshi et al., 2020; Maddipatla et al., 2023). The analysis of peripheral blood mononuclear cells (PBMC) involves studying the characteristics of these immune cells to understand how these profiles differ between diseases and healthy cohorts. Looking at differences in immune profiles can allow researchers to identify a component of the multifactorial causes of Crohn's disease and lead to potential interventions (Brown et al., 2024).

In recent years, the plastic phenotype of different immune cells has been recognized as an important player in not only eliciting an appropriate immune response, but also in the development of distinct pathologies, suggesting that there is not a single pathological landscape of Crohn's disease (Ueno et al., 2015; Wu et al., 2020). Thus, biologics can be used to manage Crohn's disease in a stepwise escalation depending on what the drug is designed to target (White et al., 2019). Treatments focus on four main categories of activation: anti-tumor necrosis factor agents (infliximab / Remicade), integrin receptor blockers (vedolizumab / Entivyo), interleukin inhibitors (risankizumab / Skyrizi), and more recently JAK inhibitors (upadacitinib / Rinvoq) (Meissner & Rao, 2025). The development of these biologics was revolutionary for the care and management of Crohn's, but many patients face primary or secondary treatment failure on these biologics. Therapies become less effective after treatment failure, with primary non-responders being 24% less likely to show remission with a different biologic (Jefremow & Neurath, 2023). Therefore, it is imperative to analyze the genetic profile of the individuals presenting for a Crohn's Disease diagnosis in order to more effectively assign them to a treatment plan.

Two crucial components of multi-omic measurement are chromatin accessibility and genetic expression, which often employ single nucleus ATAC-seq (snATAC-seq) and single

nucleus RNA-seq (snRNA-seq) data analysis respectively. The assay for transposase-accessible chromatin using sequencing (ATAC-seq) method is used in high-throughput sequencing as a scalable way to “detect the unique chromatin landscape associated with a cell type and how it may be altered by perturbation or disease” (Grandi et al., 2022). ATAC-seq is particularly useful because it does not require a large number of input cells nor a priori knowledge of transcription factors or epigenetic markers of genes of interest. The methodology of ATAC-seq is also relatively simple; a hyperactive Tn5 transposase enzyme is preloaded with sequencing adapters to determine sites of accessible chromatin (Grandi et al., 2022). Analyzing the data that is produced by ATAC-seq methodology can provide insights into mapping nucleosomes, identifying transcription factor binding sites, and measuring differential activity of regulatory elements in DNA (Smith & Sheffield, 2020). ATAC-seq analysis is inexpensive, and although protocols are constantly evolving, it generally follows a set workflow. Quality checks are completed on the dataset, adapters are trimmed, and reads are aligned to a reference genome. The results produced from ATAC-seq are generated into peaks that correlate with regions of interest (Smith & Sheffield, 2020). Using ATAC-seq, it is possible to identify regions of the genome that are bound by this enzyme and compare these regions across cohorts to determine differences in gene expression (Grandi et al., 2022).

RNA sequencing analyzes the transcriptome of a sample to reveal differential expression of a set of genes in the sample. snRNA-seq came to be as an alternative to single cell RNA-sequencing in cases where isolating the nucleus of a given cell type is easier than to isolate the cell (Kim et al., 2023). Because ATAC-seq requires cells to be lysed for access to the nucleus, snRNA-seq is crucial for multi-omic analysis. snRNA-seq begins with nuclear dissociation to obtain nuclei without damaging the membranes, then separates nuclei using centrifugation, and finally processes the nuclear RNA into cDNA libraries (Illumina, 2025). After processing, an expression matrix yielding counts of each gene in each cell, as denoted by the cell barcode, is necessary for analyses such as cell clustering.

This study aims to characterize the genomic landscapes from PBMC profiles of IBD patients and how they may advance understanding of effective IBD treatments upon diagnosis. We hypothesize that PBMCs from IBD patients will show enrichment in genes and pathways involved in inflammation compared to healthy controls.

METHODS

Data was collected via the National Library of Medicine National Center for Biotechnology Information SRA Lookup by using the following prompt to search for IBD related

studies with ATAC-seq data: ("ATAC-seq"[Strategy] OR "ATAC-seq"[Technique]) AND ("Inflammatory Bowel Disease"[MeSH Terms] OR "Inflammatory Bowel Disease"[Title/Abstract] OR "IBD"[Title/Abstract] OR "Crohn's Disease"[Title/Abstract] OR "Ulcerative Colitis"[Title/Abstract]) AND "Homo sapiens"[Organism].

The dataset selected contains raw single nucleus ATAC-seq data and 10x Genomics snRNA-seq data from a study on peripheral blood mononuclear cells (PBMCs) in a cohort of African American Crohn's Disease patients (Brown et al., 2024; Georgia Institute of Technology, 2024). The patient demographics included both male and female patients between the ages of 4 - 64. Patients either were healthy or had been diagnosed with Crohn's Disease. For those who were diagnosed, data was collected on the type of medication they were taking, their response status to the medication, their disease phenotype, if a surgery was needed to specifically address their disease, and the length of remission if they were in remission. Two multi-ome read groups were used for this analysis: Pool 3 and Pool 4. Pool 3 included three healthy individuals, serving as a control, and Pool 4 featured two Crohn's disease patients, comprising the disease group. These two pools contained a diverse range of demographics from the study, including both males and females, young and old patients, and patients who had surgery alongside those who did not.

Sample 1, Pool 3, and Pool 4 read groups (as referred to in the repository) Gene Expression Omnibus accession correspond to series GSE244831, samples GSM7830516, GSM7830519 and GSM7830520 for sn-RNAseq, and GSM7830529, GSM7830532, GSM7830533 for sn-ATACseq. Sample 1 and Pool 3 included reads distributed into 8 fastq files, whereas Pool 4 reads spanned two files. For the purpose of analysis, Pool 4 files were divided into 4 files each using seqkit. The data is multiomic homosapien data prepared with Chromium Next GEM. The snATAC-seq and snRNA-seq experiment was performed on the Illumina NGS platform, and was sequenced using paired-ends for snATAC-seq and single-end for snRNA-seq (Brown et al., 2024; Georgia Institute of Technology, 2024).

To ensure a standardized and replicable approach, this snATAC-seq analysis follows the ATAC-seq portion explored in the class BIOINF545 at the University of Michigan; a full guideline can be found at the following website: <https://theparkerlab.med.umich.edu/data/porchard/2025-03-31-multiome-lab/multiome-lab.html> (ParkerLab, 2017/2025). Raw split fastaq files from the snATAC-seq components of the data were downloaded through SRA tool kit. Hg19 was selected as the reference genome for all alignment steps. snATAC-seq reads were processed using the snATACseq-NextFlow pipeline integrating alignment and quality control mounted by Peter Porchad, which can be found using the following link:

<https://github.com/porchard/snATACseq-NextFlow.git> (Orchard, 2019/2025). The pipeline includes adaptor trimming, fastQC analysis of files at multiple stages (pre and post-trimming, after merging read groups, nucleus barcode correction, alignment of reads to reference genome using BWA, calculation of multiple QV metrics returned by *ataqv*, and peak calling in pseudo bulk results using MACS2. However, snATAC-seq was only completed on Pool 3, and computing resources were not sufficient to finish processing Sample 1 and Pool 4 concurrently.

Single-nucleus RNA sequencing data was obtained from peripheral blood mononuclear cells (PBMCs) in two datasets: Pool 3, composed of healthy individuals, and Pool 4, composed of individuals diagnosed with Crohn's disease. snRNA-seq data was analyzed using R v4.5. Sample quality was ensured by removing cells with fewer than 500 total counts, and genes expressed in fewer than three cells were excluded. Although a minimum of 1000 total counts to keep a gene is common in scRNA-seq analysis, the threshold of 500 is favorable for snRNA-seq because, in only maintaining RNA from a cell's nucleus, snRNA-seq inherently yields fewer transcript counts (Bakken et al., 2018). This preprocessing step was aimed at reducing noise in the sample and improving the quality of subsequent downstream analysis. To adjust for sequencing depth and potential technical variations in the set, the data was normalized using a scale factor of 10,000. This step ensures that differences observed between samples are more likely biological than technical. Post-normalization, highly variable features were identified using the variance - stabilizing transformation method of the Seurat v5.2.1 package, highlighting biologically significant heterogeneities. Principal component analysis (PCA) was performed on the normalized data to reduce dimensionality and simplify the complexity of the dataset while preserving its inherent variability. The analysis was conducted with 50 components to capture the major sources of variation. Uniform Manifold Approximation and Projection (UMAP) was applied to the principal components to visualize the data in a two dimensional space, facilitating the identification of distinct cell populations. SingleR functions were utilized to annotate cell types based on a reference genome from the *HumanPrimaryCellAtlasData*, ensuring accurate classification of cell types according to established transcriptional profiles (Mabbott et al., 2013; Aran et al., 2019). Cells were clustered using a graph-based method with a resolution of 0.5 for both datasets. Marker genes for clusters were identified to find differentially expressed genes. The following genes commonly identified with Crohn's Disease and targeted with biologics were the primary focus of gene enrichment and pathway analysis: NOD2, IL23 (Skyrizi), JAK 1 and 3 (Rinvoq), TNF- α (Remicade), and α 4 β 7 integrin (Entivyo) (Annese, 2020). IL23 and α 4 β 7 were found to be missing from the dataset, and were subsequently excluded from analysis. However, similar genes were incorporated in order to ensure similarity with the current treatment options

available. Although IL12 is more similar to IL23 than IL27, IL12 was also not present in the dataset. Therefore, IL27 was substituted for IL23 and $\alpha 4\beta 7$ was substituted for ITGA4. Additionally, CD4 was included to encompass possible relationships with T cells. Gene markers were annotated using the *org.Hs.eg.db* of the Bioconductor 3.21 module. EntrezIDs were mapped for each gene using the *org.Hs.eg.db* package to facilitate pathway analysis. Differential expression was calculated and subsequently ranked by log2 fold-change values for downstream analysis. Enrichment analysis of biological pathways was conducted using gene set enrichment function *gseGO* from the clusterProfiler package (Korotkevich et al., 2019; Xu et al., 2024). This analysis centered on Gene Ontology (GO) terms related to biological processes, aiming to differentiate the pathways that were distinctly active or repressed in Crohn's disease compared to healthy controls. Finally, plots were created using ggplot2 to visualize normalized enrichment scores of top enriched pathways, identifying differentially expressed genes, and comparing across selected genes unique to biologic treatments.

RESULTS

After processing, the *ataqv* analysis of the pseudo bulk snATAC-seq results showed that the mean MAPQ value for all samples was over 50,000. TSS enrichment score was over 4, and over 90% of the reads were uniquely mapped to the genome. Moreover, less than 0.1% of reads were mapped to the mitochondrial genome. For snRNA-seq, the fraction of bases with Q30 was over 88% for all reads over all samples, indicating a high confidence in base calling, and the fraction of reads mapped confidently to the transcriptome was between 62% and 67%, whereas over 97% mapped to the whole genome. Additionally, over 95% of the barcodes could be resolved properly. Together, these metrics indicate that the reads are of high quality and suitable for analysis. **Table 1** highlights the quality control metrics for snRNA-seq analysis.

Table 1. QC metrics for snRNA-seq reads

Sample	UMI Q30	Barcode Q30	Read 2 Q30	Sequenced Read Pairs	Reads Mapped Confidently to Transcriptome
Sample_1	0.9594	0.9615	0.8929	83658719	0.6686
Pool_3	0.9588	0.9696	0.9005	69818937	0.6267
Pool_4	0.8861	0.9393	0.9188	352475093	0.6243

Figure 1 shows the comparison of the UMAP clustering results between the control and disease cohorts, with annotations added using the SingleR package as compared to the reference genome *HumanPrimaryCellAtlasData* (Mabbott et al., 2013; Aran et al., 2019). These results revealed distinct clusters for both control and disease samples that effectively resolve hematopoietic lineages regardless of the mixed nature of the samples. Between control and disease state, it seems the control Pool 3 shows a more heterogeneous myeloid cell population, as the cluster is split in three. However, this could be a product of cells from distinct individuals aggregating together. Clustering data facilitated identification of cell populations specific to disease states.

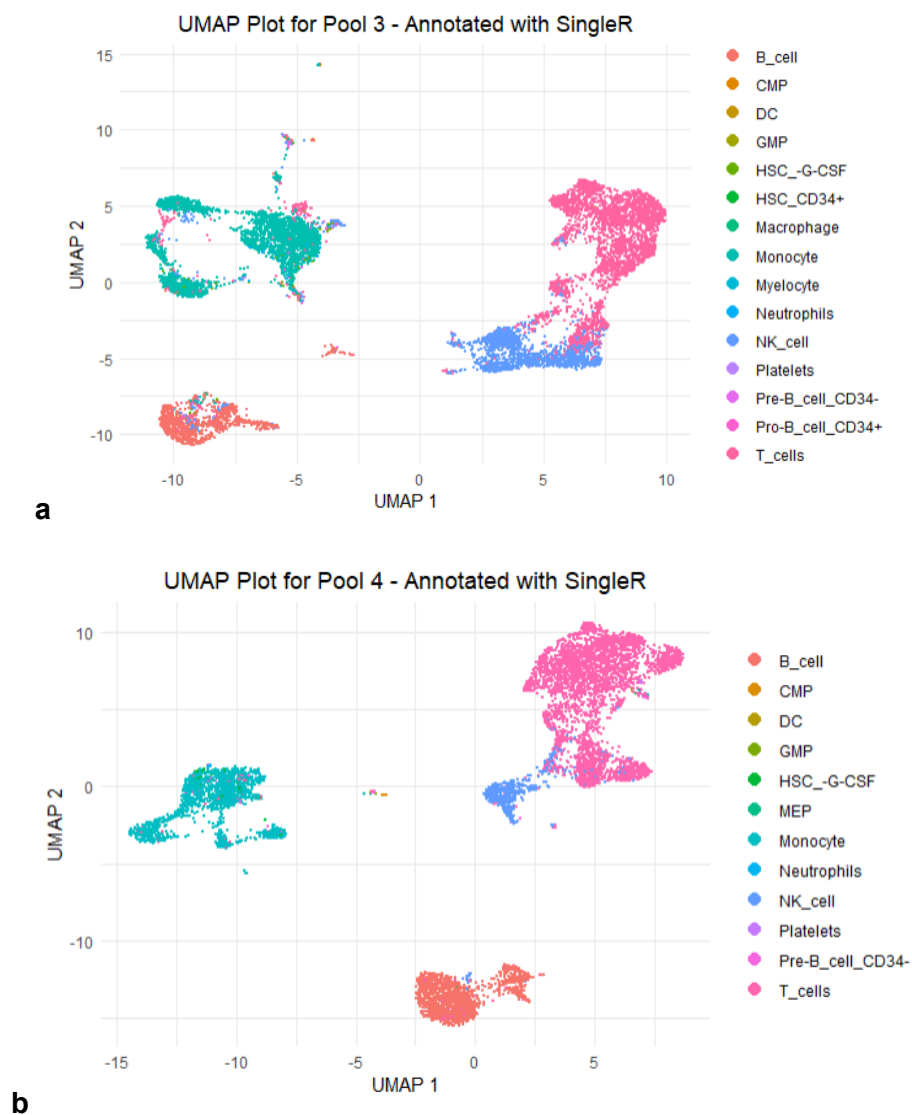


Figure 1: UMAP clustering results from the control (**a**) and disease (**b**) cohorts.

Figure 2 shows heatmaps of the top 10 differentially expressed genes between each cluster for both the disease and control states. Differential expression analysis highlighted top marker genes in each condition with the disease state showing altered expression in genes involved in immune response and cell proliferation. Many of the top 10 genes identified in control states are involved with the nervous system, normal cell physiology, and inflammation. While this set of genes is enriched mainly in the B cell population, and somewhat in the monocyte population, it is downregulated in the rest of the cell populations. In contrast, the diseased samples showed an enrichment for a different set of genes, as well as a loss of downregulation in cell populations outside of B cells. Many of the top genes were associated with lymphocyte development (LEF1, TCF7), B cell activation (PRKCA), and cell migration and immune synapse formation (ARGHAP15). Surprisingly, an enrichment for these genes was not observed in the T cell population, especially as BCL11B is a crucial, lineage-defining marker for T cells. The expression of BCL11B on B cells instead suggests an abnormal state and potentially transformed lineage. These results suggest that IBD may be accompanied with a loss of downregulation of genes in immune cell populations, and the upregulation of lymphocyte development genes.

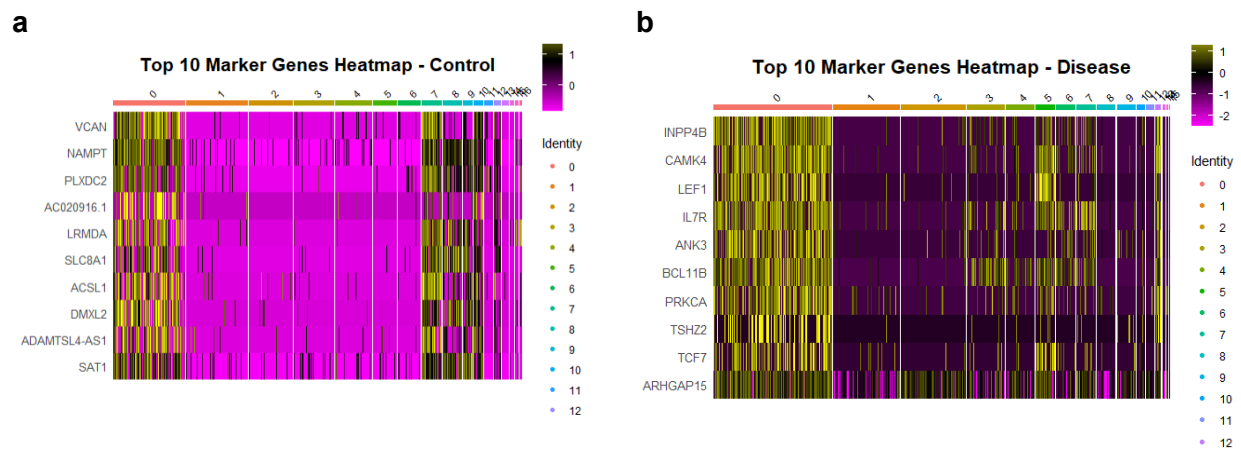


Figure 2: Heatmap of top 10 marker genes identified in control (a) and disease (b) cohorts and their frequency in each cluster.

Figure 3 shows the top ten pathways in both cohorts, control and disease, as ranked based on their normalized enrichment scores (NES). A higher absolute value NES indicates a higher degree of enrichment, a negative NES indicates that the genes in the set are downregulated in a specific pathway, and a positive NES indicates that the genes in the set are being upregulated in the pathway of the condition. The pathways shown are statistically significant, with p-values less than the threshold of 0.01. The control condition has an

enrichment for immune response, and more importantly, immune regulation pathway. On the other hand, the disease condition is enriched for cell division and cell cycle pathways, which suggests inappropriate cell proliferation. Enrichment of cell division and cycle pathways have been seen *in vitro* using samples from IBD and Crohn's disease patients, where researchers observe T cell expansion and resistance to apoptosis and altered B cell expansion and maturation (Ina et al., 1999; Doorenspleet et al., 2017; Rabe et al., 2019; Kappel-Latif et al., 2024).

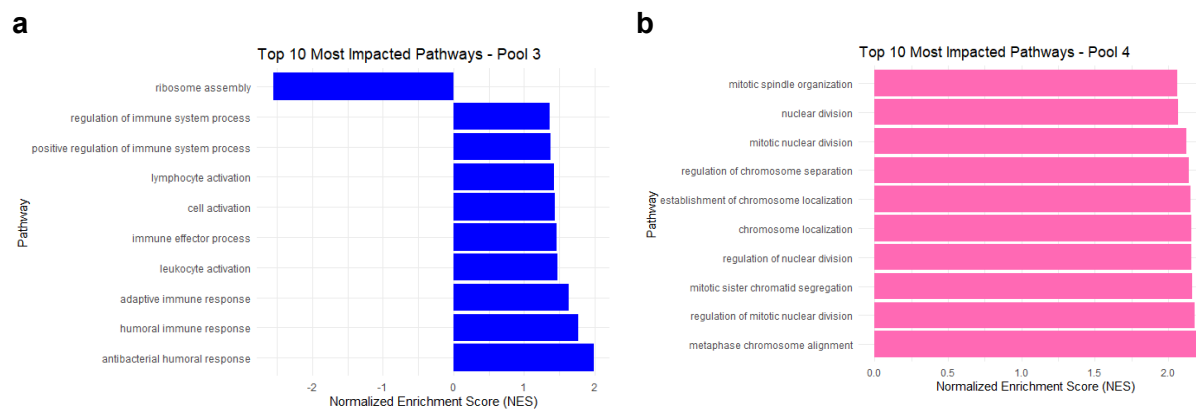


Figure 3: The highest enriched pathways in the control (a) and disease (b) cohorts and their associated NES scores.

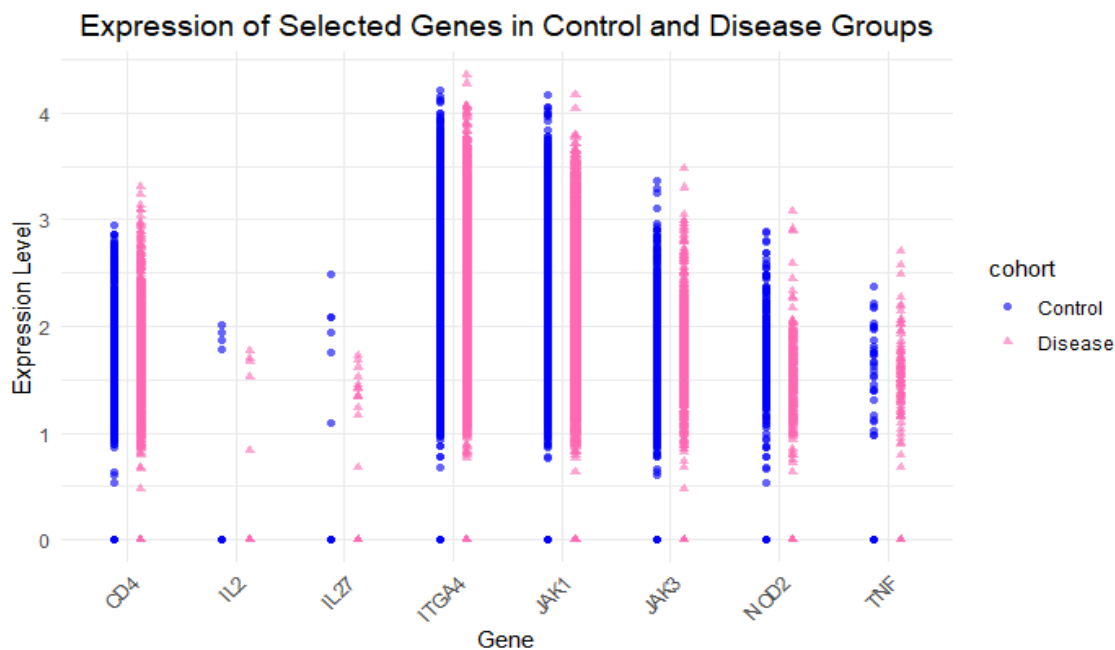


Figure 4: Normalized expression of selected genes in each cohort

Figure 4 shows the comparison of normalized expression levels of selected genes in the control and disease cohorts. Genes identified as targeted by current FDA approved medications for Crohn's disease treatment were selected and analyzed for their expression level in both cohorts (Annese, 2020). Feature analysis of the genes revealed no noticeable difference in the expression level of the genes in control and disease sample. One possible explanation for this observation is that our analysis was performed on PBMCs, which circulate in the vascular system. Thus, we may not have been able to capture differences in gene expression if the genes that are known to be overactive in Crohn's disease may only increase in expression at the relevant tissue sites of inflammation, i.e. the gastrointestinal tract, rather than systemically.

DISCUSSION

Previous work in inflammatory bowel disease highlights a clear genetic predisposition, even though the exact etiology is unknown. However, treatment options with genetic targets still fall short of a cure for the disease, with patients going through a gauntlet of treatment options before finding one that alleviates the symptoms, or in the best case scenario, prompts remission (Annese, 2020). New aspects of Crohn's Disease are constantly being investigated, including PBMCs, driving analysis of different pathways that may contribute to this condition (Brown et al., 2024). As different contributing pathways are identified, it is important that their relation to treatment options is explored as well. Therefore, our analysis investigated PBMCs and the differences in functional pathways identified between healthy and diseased cohorts by comparing genes that have a known association with Crohn's Disease and the functional pathways they affect. Main genes that have been found to be involved in IBD pathogenesis include NOD2, ITLN1, IL2, IL27, IL23, JAK 1 and 3, TNF- α , and $\alpha 4\beta 7$ integrin (Annese, 2020). In our analysis, the shift from immune-specific pathways in controls to cell cycle focused pathways in disease samples highlights a potential dysregulation that aligns with known pathological processes in Crohn's Disease where inflammation and tissue remodeling are prevalent. The differential expression analysis of the top markers found in the two cohorts showed more differences between groups than the selected genes, which is contrary to the expected result. However, it is important to note that the sample was sourced from PBMCs, not the epithelial cells of the lining of the intestines where the inflammation occurs, which might contribute to the discrepancy seen between the disease and control cohorts. Identified differentially expressed genes and enriched pathways could offer new targets for intervention, specifically addressing immune system modulation and cellular growth control. These findings

should be considered relative to existing treatments that target inflammation but might not fully address aberrant cell cycle processes uncovered in this analysis.

Previous cluster analyses on immune cells in Crohn's Disease have shown varied results. One such analysis on scRNA-seq data showed an unusual T cell landscape compared to healthy controls. Even within the T cell landscape, there were unique subsets in CD4 and CD8 clusters (Jaeger et al., 2021). In our analysis, the PBMC sample as a whole also showed some variation between diseased cohorts and healthy cohorts. Interestingly, the control group showed enrichment for pathways involved in immune system activations, whereas the disease state had enriched pathways involved in cell cycle progression. However, given the sample size, further investigation is needed to assess the validity of this finding. Another finding of note in genetic identification of Crohn's Disease is that there is an unclear relationship between genetic-based subgroups and typically used clinical parameters. Variants that have been found as possible contributors to Crohn's Disease in studies comparing healthy patients against those with diagnosed and active disease are not necessary for the disease to express itself, implying an external contributor that current genetic analysis is lacking (Cleynen et al., 2010). In the context of identifying the genetic variants associated with this disease, individual variances in states of remission or active disease might prove more useful than comparing cohort differences.

As with any novel methodology, lessons can be learned from challenges experienced during analysis. First, Crohn's Disease as a condition is highly variable and exhibits intrinsic variances, including different individuals exhibiting different phenotypes, response or non-response to medication and immunosuppressants, and diagnostic ambiguity between Ulcerative Colitis and Crohn's Disease (Liu & Anderson, 2014). Additionally, there are stages of 'flare-up' and 'remission' within the disease profiles themselves. Therefore, comparison across datasets is difficult when looking at genome wide association studies of Crohn's Disease as the population of patients varies so widely (Liu & Anderson, 2014). Second, the study team had limited experience in genetic analysis, so more complex methods that would provide a more in-depth and comprehensive analysis could not be accurately implemented. If we could redo the analysis, we could have incorporated more samples to increase the statistical power of the study. Additionally, we could have done a comparison between our identified pathways and those of previous works in the domain as a validation of our findings. Moreover, we would integrate more closely the snATAC-seq and snRNA-seq components of the multi-ome dataset, for example by implementing and refining clusters. Another aspect that would heighten the analysis would have been to recruit more advanced parallel computing power to be able to run

the snATAC-seq analysis concurrently with the snRNA-seq. Pseudo bulk peak calls from the pipeline could be run through HOMER v4.11.1 to identify enriched motifs. By applying this analysis pipeline, we could expect to identify distinct patterns of motif enrichment in the areas of open chromatin of PBMCs.

If there was sufficient time to extend the analysis, next steps would include TF footprinting of the data and a comprehensive analysis of the binding site regions to more accurately identify pathways involved in the development and symptoms of these diseases. Additionally, the study would benefit from full deconvolution of pooled samples. Future research could investigate how these findings might integrate with the development of novel therapies or diagnostics, with larger datasets potentially corroborating or extending these insights. There has also been promising work in using known loci associated with Crohn's Disease to predict those who are susceptible to acquiring the disease. Future work could include identifying patients with IBD without an official Crohn's diagnosis and analyzing their future risk by comparing their genetic profile with known Crohn's regions of interest (Liu & Anderson, 2014). Looking forward, we would also want to incorporate a more individualized analysis, given the variance in Crohn's Disease phenotypes across individuals and the uncertainty that risk alleles are indicative of a diagnosis (Cleynen et al., 2010). Finding a study where the genetic landscape was compared pre-diagnosis, post-diagnosis, and when in different stages of the disease could provide more insight into the factors that determine the cause and severity of Crohn's Disease. Establishing a comprehensive epigenomic signature associated with inflammatory responses may improve our understanding of IBD pathogenesis and guide therapeutic strategies preventing immune cell mobilization and further development of inflammation in the gut.

A major limitation of the study is the use of PBMCs without the inclusion of gut resident immune cells. Although the results obtained from circulating cells provide insights into the changes involved in IBD in a systemic context, it's impossible to directly translate them to the functional consequences deriving in inflammation. Moreover, PBMCs transcriptome is particularly susceptible to change if the donor was undergoing an infection. Another limitation is the ethnic representation among the donors. Including a single ethnic group provides a platform for more direct comparisons, but limits the reach of the conclusions. With regards to methodology, peak calling and differential chromatin accessibility were done from bulk files without clustering according to cell types. As such, the sensitivity and specificity of detection was diminished. Additionally, samples derived from pooled donors were not deconvoluted, further reducing the ability to detect changes.

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INDIVIDUAL CONTRIBUTIONS

NIKKI WATSON is a Master's student in Industrial and Operations Engineering at the University of Michigan. She holds an undergraduate degree in Kinesiology from California Polytechnic State University, SLO. Nikki focused primarily on doing the RNAseq data analysis and was a primary contributor on the report, with special attention paid to literature review and the introduction, synthesizing the results, creating the visualizations, ensuring compliance with submission guidelines, and summarizing the main points of discussion.

CARA TEIXEIRA is a Master's student in Computational Medicine and Bioinformatics at the University of Michigan, holding an undergraduate degree in Neuroscience from the same university. In this report, Cara contributed to the project conception and to the snRNA-seq data collection, quality control, analysis, and visualization pipeline. In terms of writing, Cara contributed to the abstract, introduction, methods, and discussion. Cara also proofread and edited the report and helped to ensure compliance with submission guidelines.

LUIS TERAN-RODRIGUEZ is a PhD pre-candidate in the Program in immunology at the University of Michigan. He acquired his bachelor's degree in cellular and molecular biochemistry from the University of Texas at El Paso. Luis contributed to project conception, and downloaded and preprocessed the snATAC-seq data files and performed motif enrichment analysis. Within the report, he wrote part of the introduction, methods and discussion sections. Overall contributed with immunology related insights for the analysis of the data.

ALLISON WANG is a PhD candidate in the Program in Immunology at the University of michigan. She has a Bachelors in Science in Biochemistry and Cell Biology from Rice University. Allison contributed to the project conception, performing quality control on the samples, processing the raw snATAC-seq data files, and analyzing the results. In this report, she wrote the results section and also contributed to the introduction and methods.

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